Do Arsenal Bottle It?

A Statistical Analysis of High-Stake Game Underperformance in Elite Football

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The “Nearly” Club

A brief history of Arsenal's pain

2022-23

#		PL	W	D	L	+/-	GD	PTS
1	Manchester City	38	28	5	5	94-33	+61	89
2	Arsenal	38	26	6	6	88-43	+45	84
3	Manchester United	38	23	6	9	58-43	+15	75

2023-24

#		PL	W	D	L	+/-	GD	PTS
1	Manchester City	38	28	7	3	96-34	+62	91
2	Arsenal	38	28	5	5	91-29	+62	89
3	Liverpool	38	24	10	4	86-41	+45	82

2024-25

#		PL	W	D	L	+/-	GD	PTS
1	Liverpool	38	25	9	4	86-41	+45	84
2	Arsenal	38	20	14	4	69-34	+35	74
3	Manchester City	38	21	8	9	72-44	+28	71

2018-19

Europa League Final 2018/2019

Thu, May 30, 2019, 3:00 AM | Baki Olimpiya Stadionu | Gianluca Rocchi

Chelsea **4 - 1** Arsenal

Full time

2017-18

EFL Cup Final 2017/2018

Mon, February 26, 2018, 12:30 AM | Wembley | Craig Pawson

Arsenal **0 - 3** Manchester City

Full time

2025-26

EFL Cup Final

Mon, March 23, 12:30 AM | Wembley Stadium | Peter Bankes

Arsenal **0 - 2** Manchester City

Full time

Ref: https://www.fotmob.com/

Notice a Pattern? Is this just bad luck — or is there a pattern? That's what this project finds out.

Let's Set the Scene

Three terms you need to know before we get into the data

The Bottle

Football slang for choking under pressure. When a team is expected to perform and catastrophically fails. Arsenal fans know this feeling intimately.

High-Stakes Match

PL matches in last 6 gameweeks, Quater-Finals and higher in Domestic Cups, Derbies (Historical Rivalries) in all domestic games, Round of 16 and higher in European Cups.

xG / xGA / Possession

xG = quality of chances created.
xGA = quality of chances conceded.
Possession = % of time with the ball.
These measure HOW you played, not just the score.

The Data

How the Dataset was built

433	8	136	3
Arsenal matches 2017–25	Seasons of data	High-stakes matches	Clubs analysed

Source: Fbref (StatHead) match logs + Fotmob (xG manually fill for missing cup data)

Clubs: Arsenal (subject) • Liverpool (elite control) • Tottenham (nearly-club control)

Analysis: t-tests • Performance Drop Index (PDI) • Logistic Regression

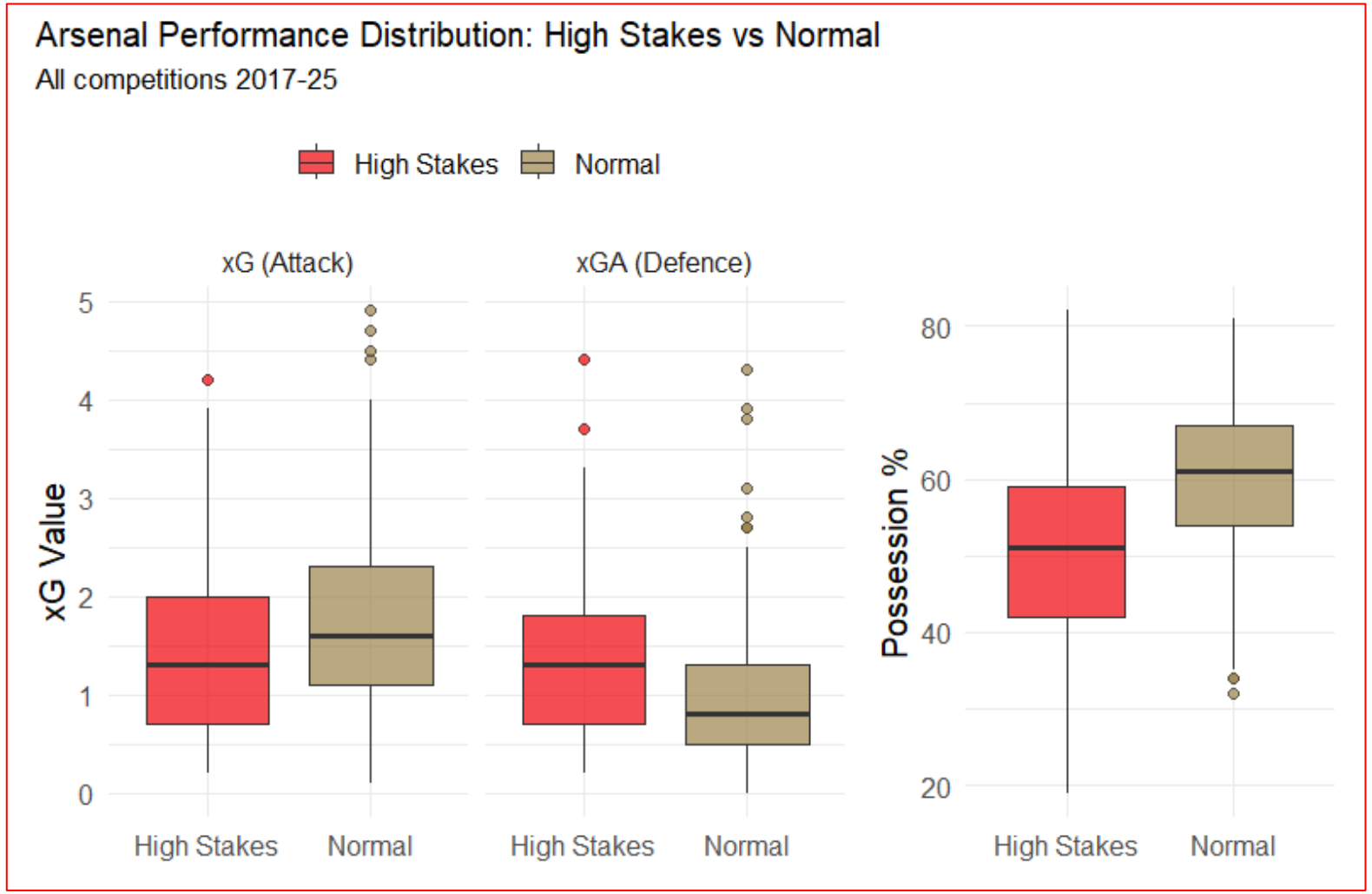
Tools: R • tidyverse • ggplot2

Q1

Do Arsenal's key metrics measurably drop in high-stakes matches?

The Numbers Don't Lie

Performance distribution + statistical significance



(Higher is better) (Lower is better) (Higher is better)

t-test Results

xG (Attack)

Normal: 1.785 → High Stake: 1.456

p = 4.7e-04 ✓

xGA (Defence)

Normal: 0.977 → High Stake: 1.355

p = 5.5e-06 ✓

Possession %

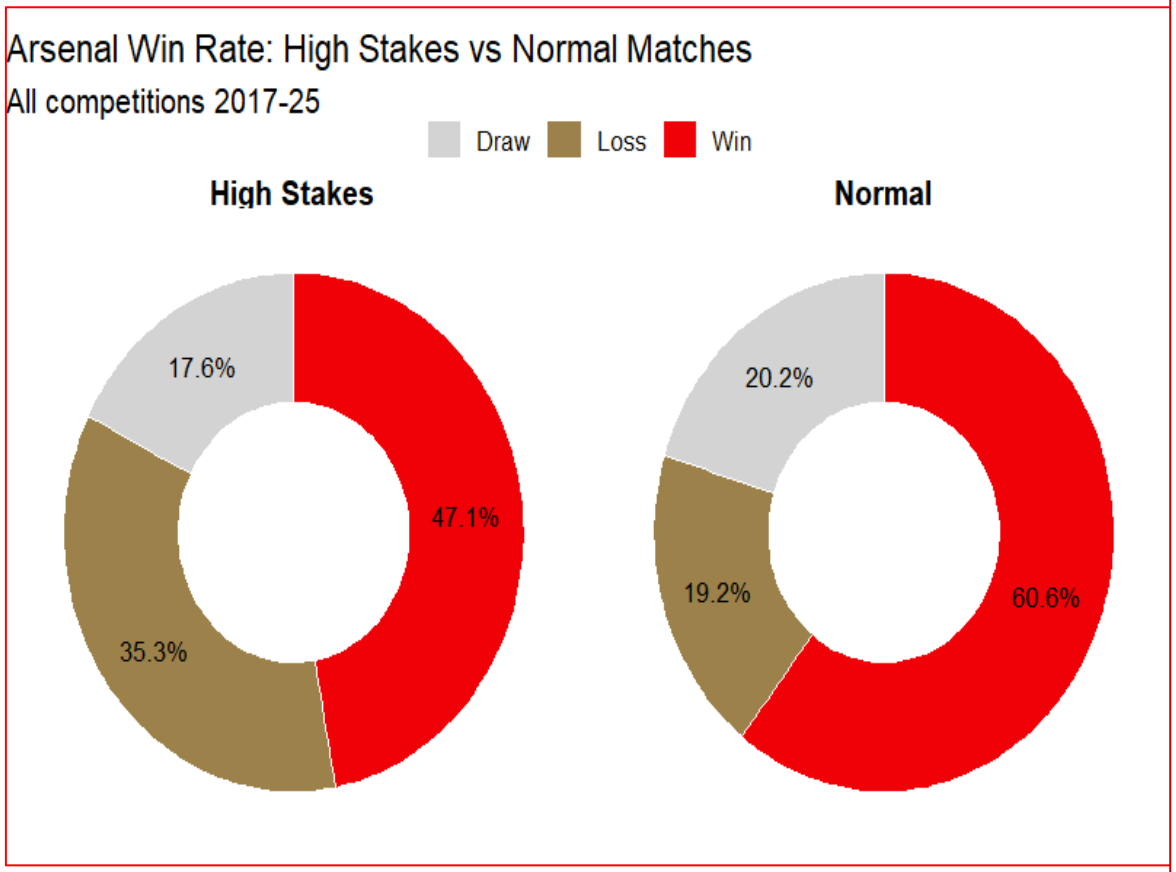
Normal: 59.7 → High Stake: 51.2

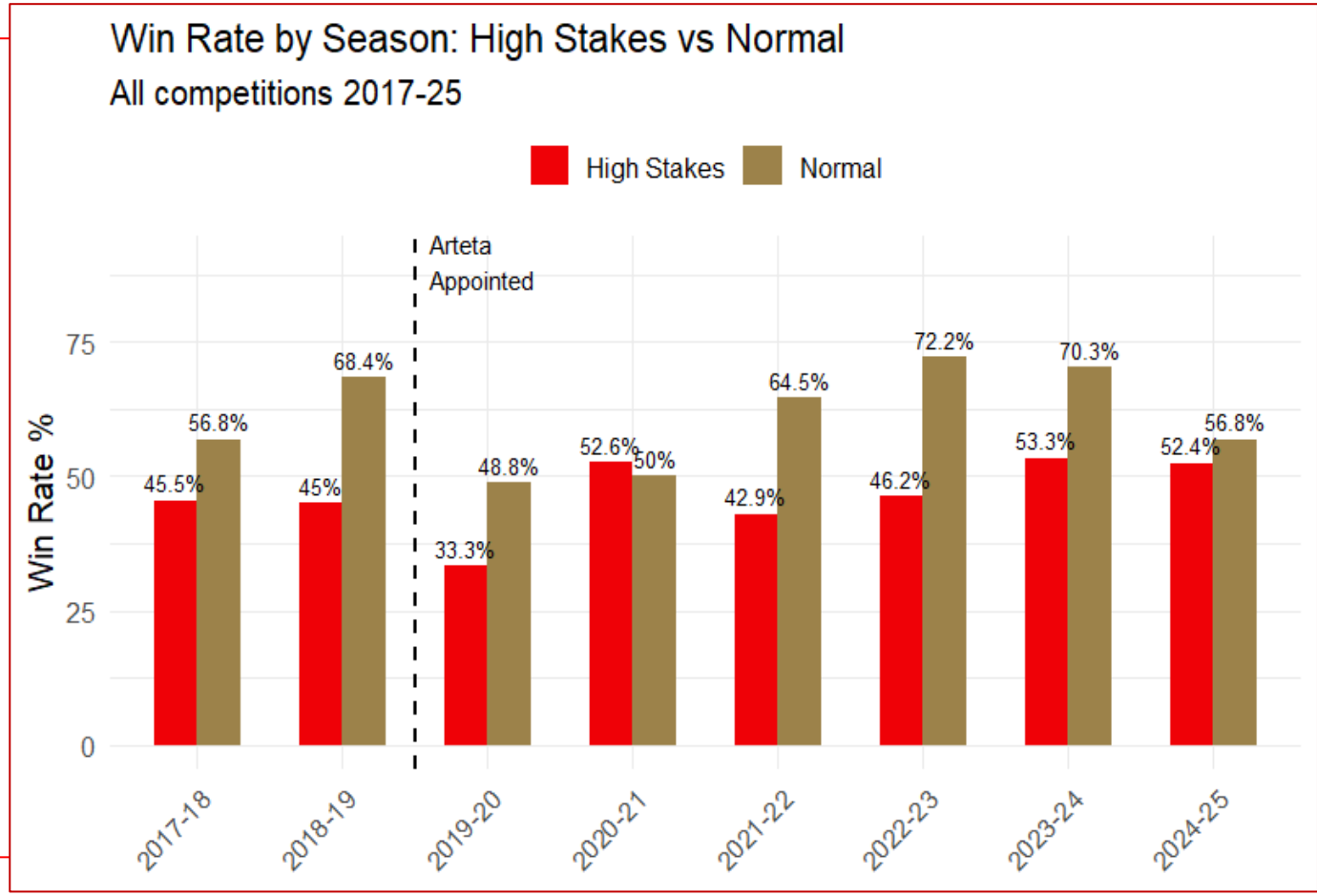
p = 3.4e-11 ✓

All three metrics statistically significant ($p < 0.05$)

60.6% → 47.1%

Win rate drops 13.5 percentage points in high-stakes matches. Loss rate nearly doubles.



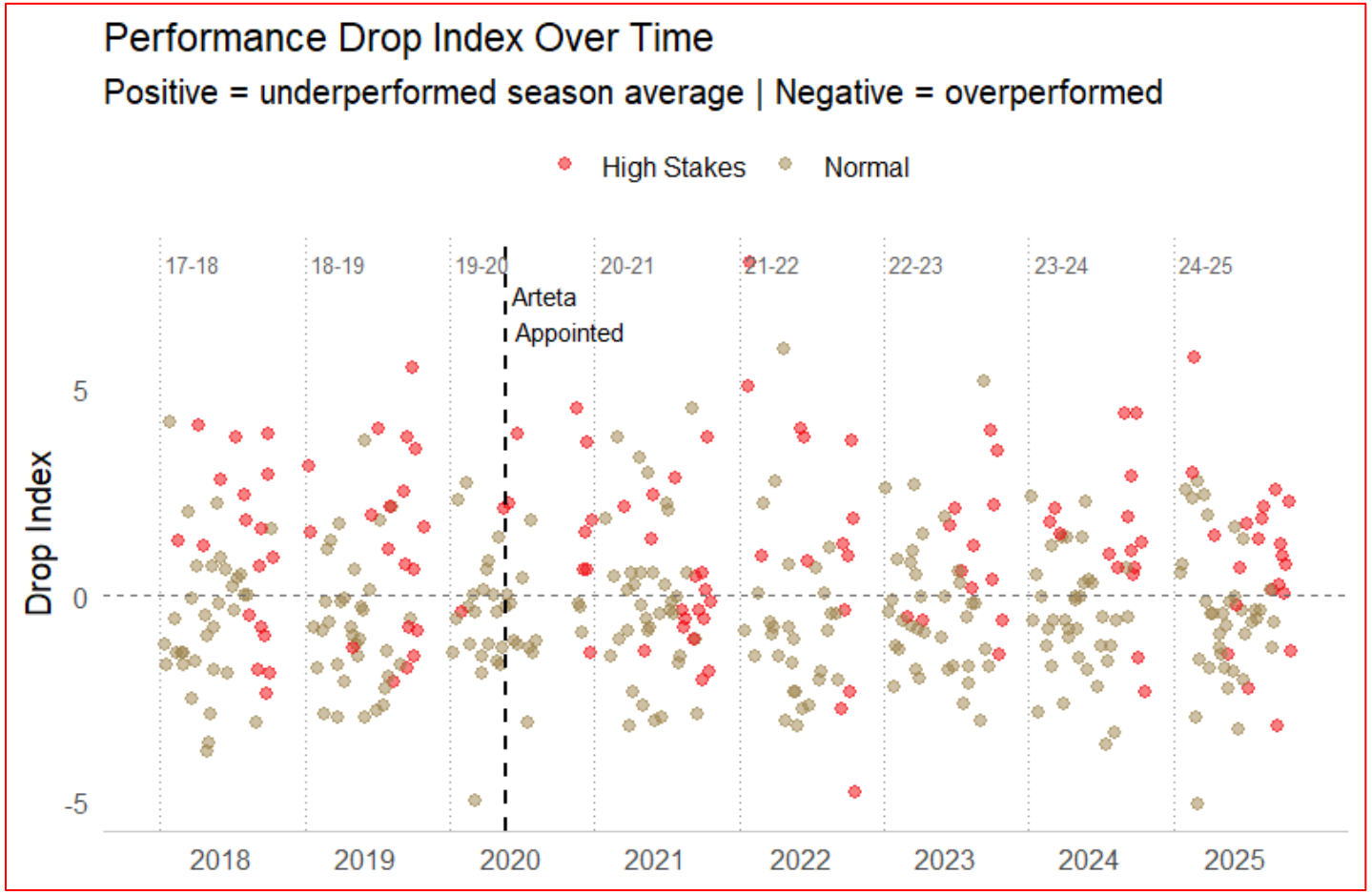


Red bar is lower than gold in 7 out of 8 seasons . This isn't a blip, it's a pattern.

Introducing: The Performance Drop Index

A composite metric combining xG drop + xGA rise + possession drop — all in one number

PDI = (season_avg_xG – match_xG) + (match_xGA - season_avg_xGA) + (season_avg_Poss – match_Poss)/10



+1.065

Average High Stakes PDI
(underperforming)

−0.494

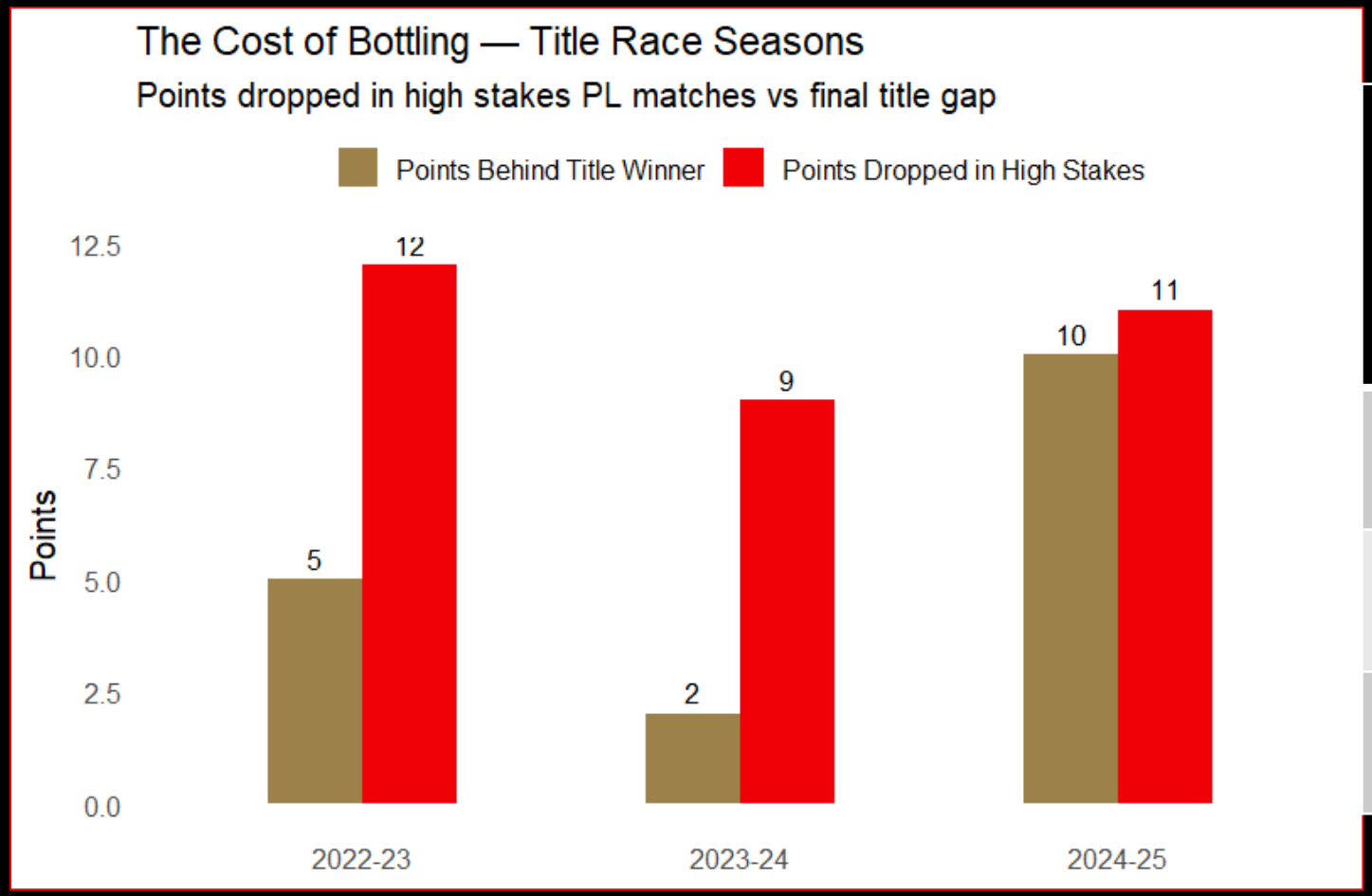
Average Normal PDI
(overperforming)

GAP: 1.559

Arsenal Overperform in a majority of their Normal Matches and Underperform in a majority of their High-Stake Matches.
Hence there is a HUGE gap in their PDI for Normal Vs High Stake.

The Almost.

Three title races. Three collapses. All the evidence.



Season	High-Stake Games	Max-Available Pts	Points Earned	Points Dropped	Lost Title By
2022-23	10	30	18	12	5
2023-24	11	33	24	9	2
2024-25	12	36	25	11	10

"They didn't lose the league to Man City.
They lost it to themselves."

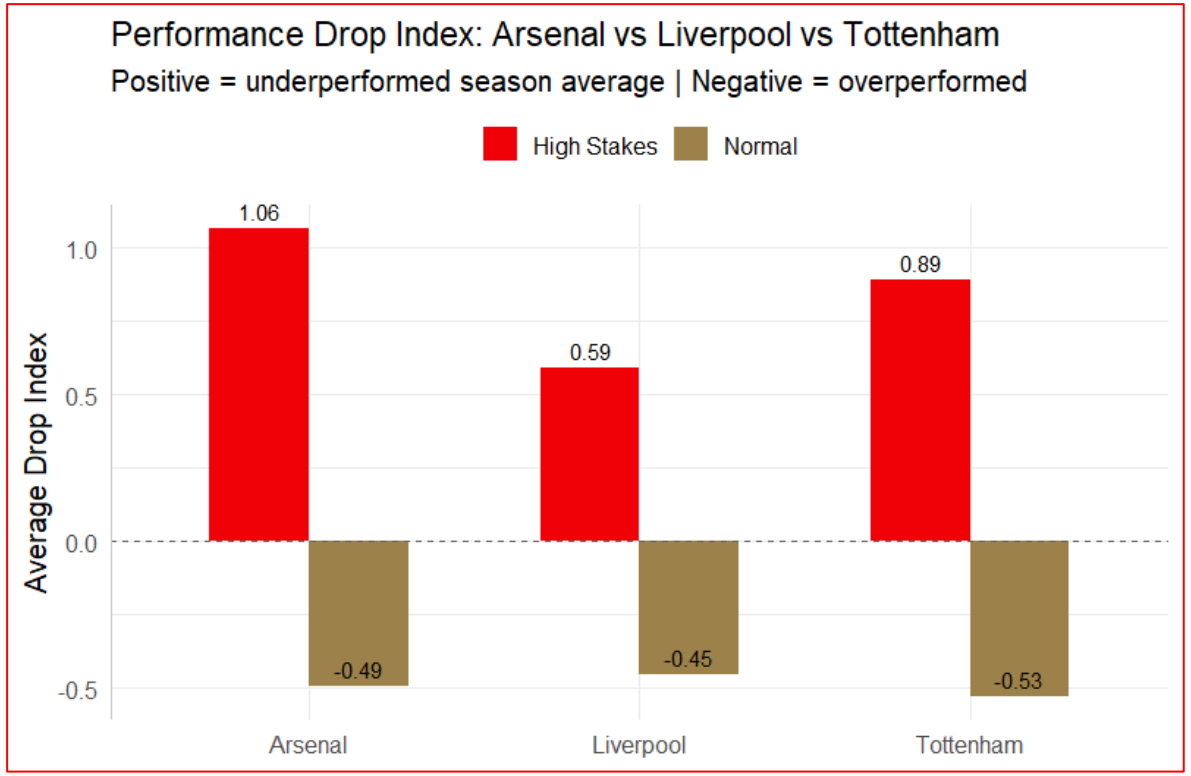
Q2

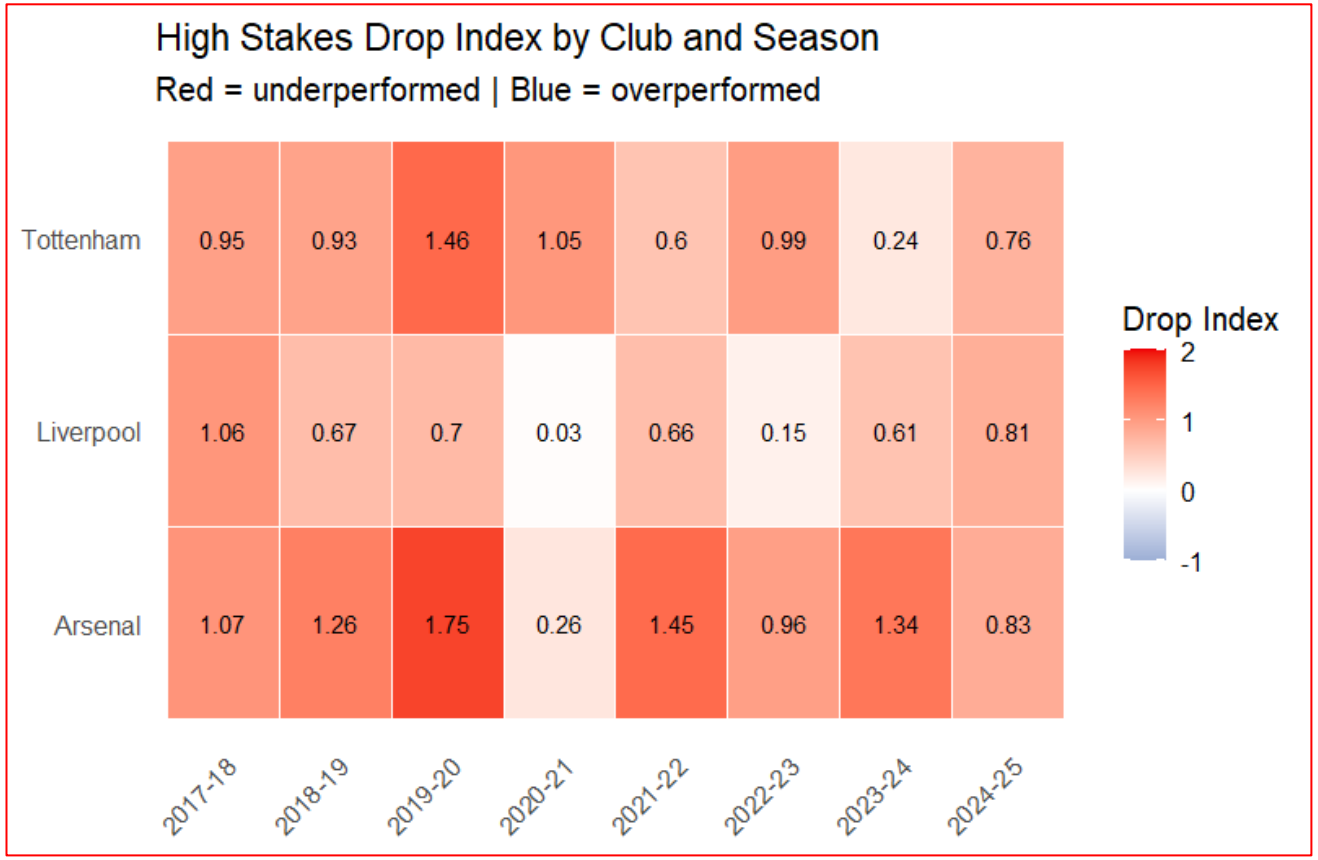
Is this unique to Arsenal — or a nearly-club disease?

Arsenal vs Liverpool vs Tottenham

Bottling: A Nearly-Club Disease

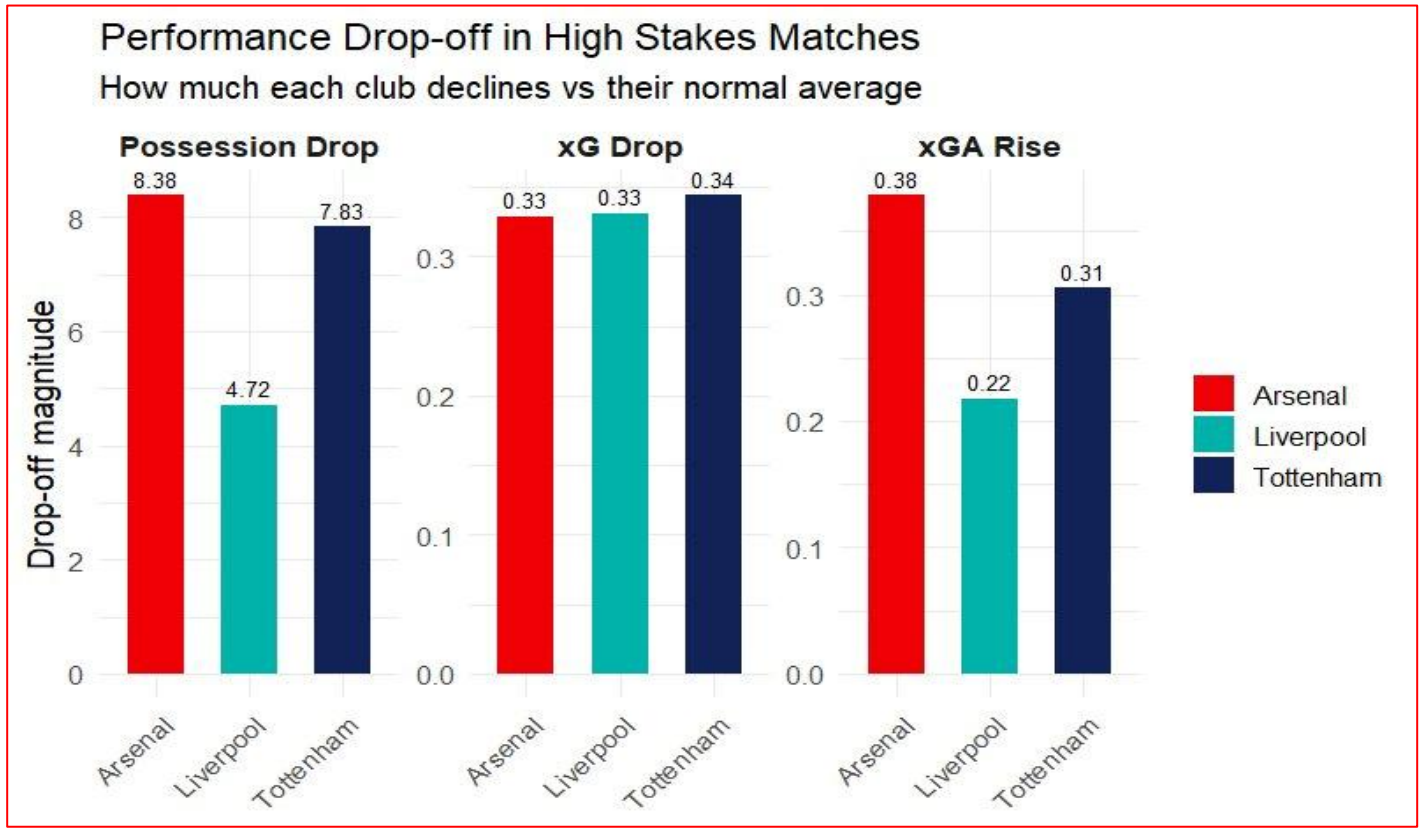
All three clubs underperform in high stakes — but Arsenal do it worst, consistently

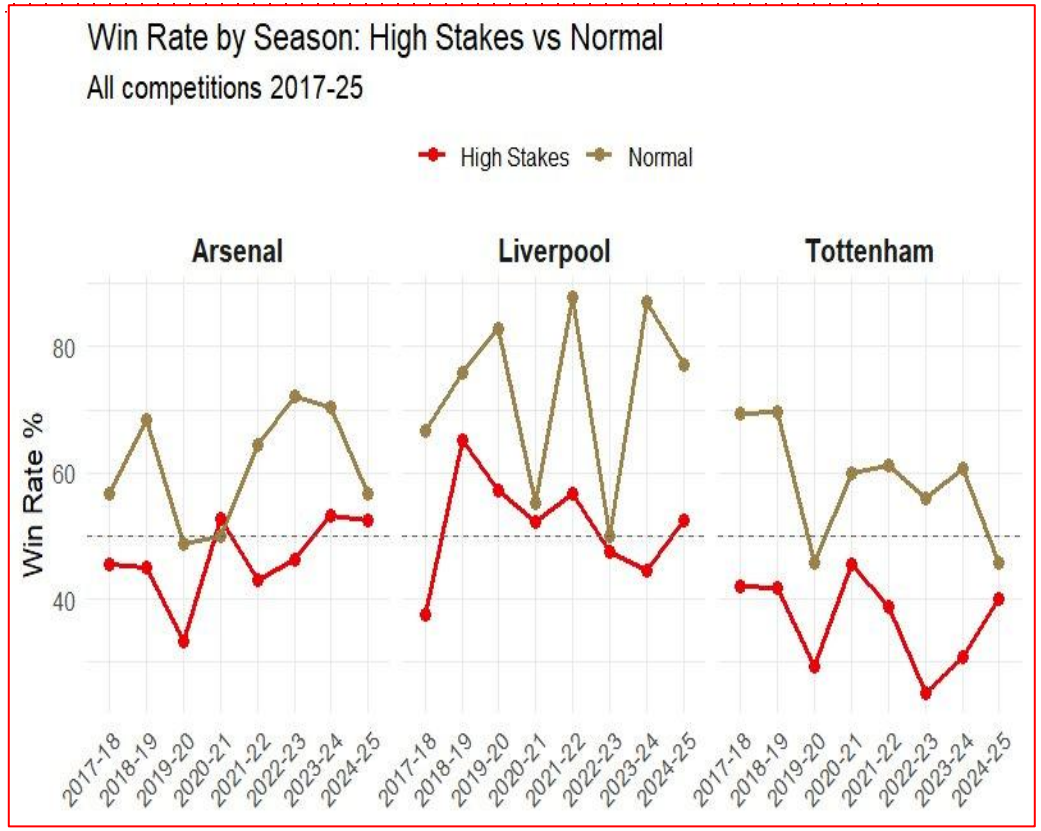




How They Bottle — And How Much

Arsenal's possession collapse is uniquely severe. Liverpool barely blink.





xG drop is universal — all clubs create less in big matches. But Arsenal and Spurs lose the ball and leak chances most.

Can We Predict the Bottle?

Logistic regression model trained on 300 Arsenal PL matches (2017–25)

High Stake	**+1.896**	***	Biggest trigger — being in a high stakes match massively increases bottle risk
Home	**−0.815**	**	Playing at home significantly reduces bottle probability
Recent Form	**−0.372**	.	Better recent form slightly reduces bottle risk
Matchweek	**−0.039**	**	Minor effect — slightly lower risk later in season

The Final Four

Arsenal's remaining 2025-26 PL fixtures — bottle probability from the model

03 May 2026	Home	MW 35	53.4%	Medium
Fulham				
10 May 2026	Away	MW 36	71.3%	High
West Ham United				
17 May 2026	Home	MW 37	51.4%	Medium
Burnley				
24 May 2026	Away	MW 38	69.7%	High
Crystal Palace				

The season ends away at Crystal Palace on the final day. The model gives that a 69.7% bottle probability.

Limitations

- xG Values, even though better than just the final scores, are not perfect
- Acquiring data was DIFFICULT. Some xG values were missing because it is a relatively new metric
- The Regression model was limited by the amount of data I had
- Football is, of course, a VERY unpredictable game

Sooo.... Do Arsenal Bottle it?

Unfortunately, Yes D: